Above every element there is a binary tree of arbitrary finite height

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Editorial note. The counted statement is the author's environment at source0.tex lines 841-846 and its proof environment opens at line 847; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

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Editorial scope. One original result of the pinned author manuscript, counted once: binary trees of arbitrary finite height above an element (author label tree3).

Allowed standard prerequisites (editorial).

- Binary tree in a Steiner triple system. A family indexed by finite binary sequences whose nodes are ordered by extension and whose elements grow with the index.. It is the object constructed.
- Products above a prescribed element. Below every decomposition of an element there are two elements above any prescribed bound.. It is the fact that lets the construction continue at each node.

Lemma 0.1 1. An HF-ordering of an infinite STS does not have a maximum.

2. If $<$ is an HF-ordering of an infinite Steiner triple system M, then for every $a \in M$, for every $b \geq a$, there are $b_1, b_2 > b$ such that $a = b_1 \cdot b_2$.

Proposition 0.2 Let $<$ be an HF-ordering of the infinite Steiner triple system M and let $n < \omega$. For every $a \in M$ there is a binary tree $T = \{a_s \mid s \in 2^{<n}\}$ in M such that

1. $a_s > a$ for every s
2. $a_s > a_t$ if $s \subsetneq t$.

Proof. We prove by induction on n that for every $a \in M$ for every finite $A \subseteq M$, there is a binary tree $T = \{a_s \mid s \in 2^{<n}\}$ in M satisfying conditions 1,2 and such that there is no block connecting elements of T with elements of A (that is, no products $t \cdot a$ with $t \in T$ and $a \in A$ are in $T \cup A$). The initial cases $n = 0, 1$ are trivial. Consider the case $n = 2$. Let A' be the set of products $\{x \cdot y \mid x, y \in A\}$, and let $b > a$ and larger than every element of A'. We use Lemma 0.1 to get some $b_1 > b_2 > b$ in M such that $b = b_1 \cdot b_2$. We let $a_\emptyset = b_1$, $a_0 = b_2$ and $a_1 = b$.

Now we assume inductively the result holds for $n \geq 2$. Given $a \in M$ and a finite set $A \subseteq M$, the inductive hypothesis gives some tree $T_0 = \{a_s \mid s \in 2^{<n}\}$ as required for a, A. Let $A' = A \cup T_0$. Again by inductive hypothesis, we can find a tree $T_1 = \{a'_s \mid s \in 2^{<n}\}$ in M with a'_s larger than every element of A', with no blocks connecting T_1 and A' and such that condition 2 holds. Now we combine T_0 and T_1 to get a binary tree $T = \{b_s \mid s \in 2^{<n+1}\}$. We set $b_{0s} = a_s$, $b_{1s} = a'_s$ and $b_\emptyset = a_\emptyset \cdot a'_\emptyset$. We first check that $a_\emptyset \cdot a'_\emptyset > a'_\emptyset$. Since $n \geq 2$ this is clear, since otherwise the blocks $\{a'_\emptyset, a'_0, a'_1\}$ and $\{a'_\emptyset, a_\emptyset, a_\emptyset \cdot a'_\emptyset\}$ are different and contradict that $<$ is an HF-ordering. Since there is no block connecting elements of T_1 with elements of T_0, T is a tree. □